

Construction of an Arbitrage-Free Local-Volatility Surface from Market Option Prices

Implied-volatility inversion, arbitrage-free interpolation, Dupire's equation,
and the model risk that survives calibration

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Abstract

We construct, from a grid of European option quotes, the unique one-dimensional diffusion consistent with every observed price — Dupire's local-volatility model — and we do so end to end: implied-volatility inversion, arbitrage-free surface construction, extraction of σ_{loc} via Dupire's equation, Monte-Carlo simulation, and exotic pricing. The mathematical heart of the exercise is that Dupire's formula requires $\partial_T C$, $\partial_K C$ and $\partial_{KK}^2 C$, while C is observed only on a coarse grid and only up to a bid–ask spread. Numerical differentiation of noisy data is a mildly ill-posed inverse problem in the sense of Hadamard: we show that the second difference has root-mean-square error $ah^2 + bh^{-2}$ with $b = \sqrt{6}\varepsilon$, so the best attainable accuracy scales only as $\varepsilon^{1/2}$ and the optimal step is *wider than the listed strike spacing*. No stencil rescues the estimator; regularisation is mandatory. We regularise by projecting the quotes onto the eSSVI family, for which the absence of butterfly arbitrage has a closed-form characterisation, and we interpolate in maturity with a monotonicity-preserving scheme so that calendar-spread arbitrage is excluded structurally rather than checked ex post. A central identity organises the whole construction: the denominator of Dupire's formula, written in implied-variance coordinates, *is* Durrleman's function, so the local variance is well defined exactly on butterfly-free slices and non-negative exactly on calendar-free surfaces. Because our synthetic market is generated by a Heston model, the true local-volatility surface is computable and our estimator can be scored against it. Finally, we show that local volatility, Heston and flat Black–Scholes — three models agreeing on every European option — disagree materially on barriers, cliquets and lookbacks, which is the empirical content of Gyöngy's mimicking theorem.

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1 Introduction

The Black–Scholes model assigns one volatility to one underlying. The market assigns a different implied volatility to every strike and every maturity. The resulting two-dimensional object, the implied-volatility surface, is not a model: it is a quotation convention, a bijective re-encoding of prices through a formula that everyone agrees to use and nobody believes. The question that opens this report is the natural one. Given the surface, is there a model?

Dupire (1994) answered it in the affirmative and with remarkable economy. If European call prices are known for a continuum of strikes and maturities, there is exactly one diffusion

$$dS_t = (r - q) S_t dt + \sigma_{\text{loc}}(S_t, t) S_t dW_t \quad (1)$$

that reproduces all of them, and its diffusion coefficient is given in closed form by what is now called Dupire’s equation,

$$\frac{\partial C}{\partial T} = \frac{1}{2} \sigma_{\text{loc}}^2(K, T) K^2 \frac{\partial^2 C}{\partial K^2} - (r - q) K \frac{\partial C}{\partial K} + qC. \quad (2)$$

This is a striking result: a complete calibration, with no optimisation, no parameters and no iteration. It is also, taken literally, unusable. The data are not a continuum; they are a few dozen numbers per expiry, quoted on a discrete strike ladder, rounded to a tick, and bracketed by a bid–ask spread. Formula (2) asks us to divide a first derivative in maturity by a second derivative in strike, both estimated from that. The second derivative in particular is a small difference of large numbers; quoting noise of one cent on a strike ladder spaced at 2.5 points enters the estimate multiplied by $h^{-2} \approx 0.16$, which is comparable to the quantity being estimated.

The practical consequence is not that the answer is slightly noisy. It is that the answer is frequently *inadmissible*: the estimated $\partial_{KK}^2 C$ changes sign, the implied risk-neutral density becomes negative, and the local variance returned by (2) is negative or explosive. Section 5 documents this on our data.

The resolution has been understood since Gatheral (2006) and is by now standard on derivatives desks, but it is worth stating as a principle rather than a recipe. *One does not differentiate data; one differentiates a model of the data.* The intermediate object must be smooth enough to differentiate twice and must be free of static arbitrage, because — and this is the identity around which the present report is organised — the denominator of Dupire’s formula written in implied-variance coordinates is precisely Durrleman’s function, whose non-negativity *is* the absence of butterfly arbitrage. Arbitrage-freeness is not a cosmetic property of the fitted surface. It is the well-posedness condition of the extraction.

What this report does

1. Sections 2–3 set up the implied-volatility transform, prove that it is well posed but ill conditioned in the wings, and characterise static arbitrage through Breeden–Litzenberger and Durrleman.
2. Section 4 proves Dupire’s theorem from the Fokker–Planck equation, derives Gatheral’s implied-variance form, and proves the identity linking its denominator to Durrleman’s function. Gyöngy’s mimicking theorem is recalled because it explains precisely what local volatility does *not* determine.
3. Section 5 quantifies the ill-posedness: an exact bias/variance decomposition of the second difference, the optimal step, and the $\varepsilon^{1/2}$ barrier.
4. Section 6 builds the regularised surface (SVI, SSVI, eSSVI), including a lemma showing that monotone interpolation of total variance excludes calendar arbitrage by construction — a step we found to be the one most easily botched.
5. Sections 7–8 give the numerical implementation and the results: fit quality, arbitrage audit, recovery of the true local-volatility surface, Monte-Carlo verification that the extracted surface reprices the vanillas, exotic pricing, and a noise-propagation study answering the question “how much of this surface is real?”

On the data

A report of this kind should ideally use a real quote file. We use instead a synthetic market generated by a Heston model and then deliberately degraded — bid–ask spread, tick rounding, vega-proportional quoting error, out-of-the-money quotes only, within a maturity-dependent moneyness band. This is not a compromise but a methodological choice: because the data-generating process is known, the *true* local-volatility surface can be computed to machine accuracy and our estimator can be scored against it. With real data one can only check self-consistency; here we can measure error. Every stage downstream of the market construction sees nothing but (K, T, mid) .

2 Preliminaries

2.1 Market and notation

We fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{Q})$ with $\mathbb{Q}$ a risk-neutral measure, a constant risk-free rate r and a constant continuous dividend yield q . The underlying S is a positive semimartingale with $S_0 > 0$; the forward for maturity T is $F_T = S_0 e^{(r-q)T}$. We write $C(K, T) = e^{-rT} \mathbb{E}^\mathbb{Q}[(S_T - K)^+]$ for the undiscounted-market call price, and throughout

$$y = \log\left(\frac{K}{F_T}\right), \quad w(y, T) = \sigma_{\text{BS}}^2(y, T) T \quad (3)$$

for the log-moneyness and the *total implied variance*. The choice of y rather than $\log(K/S_0)$ matters: it is the coordinate in which the smile is approximately stationary and in which the arbitrage conditions take their simplest form.

2.2 Black–Scholes and the implied-volatility transform

Under $dS_t = (r - q)S_t dt + \sigma S_t dW_t$ the call price is

$$C^{\text{BS}}(S, K, T, \sigma) = S e^{-qT} \mathcal{N}(d_+) - K e^{-rT} \mathcal{N}(d_-), \quad d_{\pm} = \frac{\log(S/K) + (r - q \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad (4)$$

with $\mathcal{N}$ the standard normal distribution function and ϕ its density. The vega is

$$\mathcal{V}(\sigma) = \frac{\partial C^{\text{BS}}}{\partial \sigma} = S e^{-qT} \sqrt{T} \phi(d_+) > 0. \quad (5)$$

Proposition 2.1 (Well-posedness of the inversion). *Let $T > 0$, $K > 0$ and let C satisfy the static no-arbitrage bounds*

$$e^{-rT}(F_T - K)^+ < C < S_0 e^{-qT}. \quad (6)$$

Then there exists a unique $\sigma \in (0, \infty)$ with $C^{\text{BS}}(S_0, K, T, \sigma) = C$. If C violates (6), no such σ exists.

Proof. By (5), $\sigma \mapsto C^{\text{BS}}(\sigma)$ is continuous and strictly increasing on $(0, \infty)$. As $\sigma \downarrow 0$ we have $d_{\pm} \rightarrow \pm\infty \cdot \text{sign}(\log(F_T/K))$, so $C^{\text{BS}} \rightarrow e^{-rT}(F_T - K)^+$; as $\sigma \uparrow \infty$, $d_+ \rightarrow +\infty$ and $d_- \rightarrow -\infty$, so $C^{\text{BS}} \rightarrow S_0 e^{-qT}$. The image of $(0, \infty)$ is therefore exactly the open interval in (6). Existence follows from the intermediate value theorem and uniqueness from strict monotonicity. The converse is immediate since the image is exactly that interval. $\square$

Proposition 2.1 is what licenses treating σ_{BS} as a *coordinate* rather than a modelling assumption: quoting a price and quoting an implied volatility are the same act.

Remark 2.2 (Conditioning in the wings). Well-posed is not the same as well conditioned. Differentiating the inverse map, a price perturbation δC produces $\delta \sigma = \delta C / \mathcal{V}(\sigma)$, and by (5)

$$\mathcal{V}(\sigma) = S_0 e^{-qT} \sqrt{T} \phi(d_+) \sim \frac{S_0 e^{-qT} \sqrt{T}}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2w}\right) \quad \text{as } |y| \rightarrow \infty,$$

so vega decays like a Gaussian in log-moneyness. A fixed tick of one cent therefore translates into an implied-volatility error that grows without bound in the wings. This is the first — and mildest — of the two amplification mechanisms in this report. By Lee (2004), admissible surfaces satisfy $w(y, T) \leq 2|y|$ for large $|y|$, so the exponent is at worst $O(|y|)$; the decay is nonetheless brutal in practice.

Numerically we invert (4) by a damped Newton iteration seeded with the at-the-money approximation of Brenner and Subrahmanyam (1988), $\sigma_0 \approx \sqrt{2\pi/T} C/S_0$, capped at half a volatility point per step, with a Brent bisection fallback on $[10^{-6}, 5]$. Quotes violating (6) are discarded rather than clipped: a quote outside the bounds is not a noisy observation of a volatility, it is not an observation of a volatility at all.

3 The implied-volatility surface and static arbitrage

3.1 Breeden–Litzenberger

Theorem 3.1 (Breeden and Litzenberger, 1978). *Let S_T have a distribution with density $p(\cdot, T)$ and suppose $\mathbb{E}[S_T] < \infty$. Then*

$$\frac{\partial C}{\partial K}(K, T) = -e^{-rT} \mathbb{Q}(S_T > K), \quad \frac{\partial^2 C}{\partial K^2}(K, T) = e^{-rT} p(K, T). \quad (7)$$

Proof. Write $C(K, T) = e^{-rT} \int_K^\infty (S - K) p(S, T) dS$. Differentiating under the integral sign (legitimate by dominated convergence, using $\mathbb{E}[S_T] < \infty$), the boundary term $(S - K)p(S, T)|_{S=K}$ vanishes and $\partial_K C = -e^{-rT} \int_K^\infty p(S, T) dS$. Differentiating once more gives $\partial_{KK}^2 C = e^{-rT} p(K, T)$. $\square$

Theorem 3.1 is the pivot of the entire subject. It says that the second derivative in strike is the risk-neutral density, hence that convexity of C in K is not a technical regularity condition but the statement that the market’s implied distribution is a distribution. It also says, less comfortably, that extracting the density from prices is an operation of *differentiation*, which is exactly the operation that does not commute with noise. We return to this in Section 5.

3.2 Static arbitrage: the two conditions

Definition 3.2 (Free of static arbitrage; Roper, 2010). A call-price surface $C(K, T)$, $K > 0$, $T \geq 0$, is free of static arbitrage if there exists a non-negative martingale S (in the discounted, dividend-adjusted sense) such that $C(K, T) = e^{-rT} \mathbb{E}[(S_T - K)^+]$ for all K, T .

Roper (2010) shows that, for a surface satisfying the obvious boundary and growth conditions, this is equivalent to the conjunction of

(B) *butterfly*: $K \mapsto C(K, T)$ convex for each T , i.e. $\partial_{KK}^2 C \geq 0$;

(C) *calendar*: $T \mapsto w(y, T)$ non-decreasing for each fixed y .

The names are the trading strategies that would otherwise be free money: a butterfly spread has non-negative payoff and would cost $\partial_{KK}^2 C dK^2 < 0$; a calendar spread of two options with the same log-moneyness would have non-positive cost and non-negative payoff.

Remark 3.3. Condition (C) is stated in *total variance* at fixed *log-moneyness*, not in implied volatility at fixed strike. Implied volatility can perfectly well decrease with maturity without arbitrage — it routinely does — while w cannot decrease. This is the reason y and w , rather than K and σ_{BS} , are the right coordinates.

3.3 Durrleman’s function

Condition (B) can be transported into implied-variance coordinates. The result is the single most useful diagnostic in this report.

Proposition 3.4 (Gatheral, 2004; Durrleman, 2010). *Let $w(\cdot, T)$ be C^2 and positive. Then $\partial_{KK}^2 C \geq 0$ for all $K > 0$ if and only if $g(y, T) \geq 0$ for all $y \in \mathbb{R}$, where*

$$g(y, T) = \left(1 - \frac{y \partial_y w}{2w}\right)^2 - \frac{(\partial_y w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{\partial_{yy}^2 w}{2}. \quad (8)$$

Moreover the implied density of y is

$$p(y, T) = \frac{g(y, T)}{\sqrt{2\pi w}} \exp\left(-\frac{d_-^2}{2}\right), \quad d_- = -\frac{y}{\sqrt{w}} - \frac{\sqrt{w}}{2}. \quad (9)$$

Proof sketch. Write the Black–Scholes price in the (y, w) parameterisation: $C = e^{-rT} F_T [\mathcal{N}(d_+) - e^y \mathcal{N}(d_-)]$ with $d_{\pm} = -y/\sqrt{w} \pm \sqrt{w}/2$. Change variables from K to y using $\partial_K = (1/K)\partial_y$, so that

$$\partial_{KK}^2 C = \frac{1}{K^2} (\partial_{yy}^2 C - \partial_y C),$$

and substitute $w = w(y, T)$, applying the chain rule to account for the dependence of $d_{\pm}$ on both y explicitly and through w . Collecting terms, all the Gaussian factors combine into $\phi(d_-)/\sqrt{w}$ and the polynomial prefactor is exactly $g(y, T)$, giving $\partial_{KK}^2 C = e^{-rT} \frac{F_T}{K^2} e^y \frac{g(y, T)}{\sqrt{2\pi w}} e^{-d_-^2/2}$. Since $F_T e^y / K^2 = 1/K$ and all remaining factors are strictly positive, $\partial_{KK}^2 C \geq 0 \iff g \geq 0$; combining with Theorem 3.1 and the Jacobian $dK = K dy$ gives (9). The detailed algebra is reproduced in Appendix A. $\square$

Equation (9) gives a cheap and stringent test of any fitted surface: the implied density must be non-negative and must integrate to one. We report both in Section 8.

4 Dupire's equation

4.1 The theorem

Theorem 4.1 (Dupire, 1994). *Suppose European call prices $C(K, T)$ are known for all $K > 0$, $T > 0$, are $C^{2,1}$ in (K, T) , and satisfy*

$$\partial_{KK}^2 C > 0 \quad \text{and} \quad \partial_T C + (r - q)K \partial_K C + qC \geq 0. \quad (10)$$

Then the diffusion (1) with

$$\sigma_{\text{loc}}^2(K, T) = \frac{\partial_T C + (r - q)K \partial_K C + qC}{\frac{1}{2}K^2 \partial_{KK}^2 C} \quad (11)$$

reproduces every one of the prices $C(K, T)$, and it is the unique one-dimensional diffusion of the form (1) that does so.

Proof. Let $p(S, T)$ denote the density of S_T under $\mathbb{Q}$. For the diffusion (1), p solves the Fokker–Planck (forward Kolmogorov) equation

$$\frac{\partial p}{\partial T} = -\frac{\partial}{\partial S} [(r - q)Sp] + \frac{1}{2} \frac{\partial^2}{\partial S^2} [\sigma_{\text{loc}}^2(S, T)S^2 p]. \quad (12)$$

Write $C(K, T) = e^{-rT} \int_K^\infty (S - K)p(S, T) dS$ and differentiate in T :

$$\frac{\partial C}{\partial T} = -rC + e^{-rT} \int_K^\infty (S - K) \frac{\partial p}{\partial T} dS. \quad (13)$$

Insert (12) and treat the two terms separately. For the drift term, one integration by parts gives

$$-\int_K^\infty (S - K) \frac{\partial}{\partial S} [(r - q)Sp] dS = \left[-(S - K)(r - q)Sp \right]_K^\infty + \int_K^\infty (r - q)Sp dS = (r - q) \int_K^\infty Sp dS,$$

the boundary terms vanishing at $S = K$ because $(S - K) = 0$ and at $S = \infty$ because $\mathbb{E}[S_T^2]$ -type integrability makes $S^2 p \rightarrow 0$ (it suffices that $\mathbb{E}[S_T] < \infty$ and p decays). Writing $S = (S - K) + K$ and using Theorem 3.1,

$$(r - q) \int_K^\infty Sp dS = (r - q) \left[e^{rT} C - K e^{rT} \partial_K C \right]. \quad (14)$$

For the diffusion term, two integrations by parts give

$$\frac{1}{2} \int_K^\infty (S - K) \frac{\partial^2}{\partial S^2} [\sigma_{\text{loc}}^2 S^2 p] dS = \frac{1}{2} \left[(S - K) \frac{\partial}{\partial S} (\sigma_{\text{loc}}^2 S^2 p) - \sigma_{\text{loc}}^2 S^2 p \right]_K^\infty + \underbrace{\frac{1}{2} [\sigma_{\text{loc}}^2 S^2 p]_{S=K}^\infty}_{\text{already included}},$$

and, collecting carefully, all boundary contributions at $S = \infty$ vanish while at $S = K$ the surviving term is $\frac{1}{2}\sigma_{\text{loc}}^2(K, T)K^2p(K, T)$:

$$\frac{1}{2} \int_K^\infty (S-K) \frac{\partial^2}{\partial S^2} [\sigma_{\text{loc}}^2 S^2 p] dS = \frac{1}{2} \sigma_{\text{loc}}^2(K, T) K^2 p(K, T). \quad (15)$$

Substituting (14) and (15) into (13), multiplying by e^{-rT} and using $p(K, T) = e^{rT} \partial_{KK}^2 C$ from Theorem 3.1:

$$\frac{\partial C}{\partial T} = -rC + (r-q)[C - K\partial_K C] + \frac{1}{2}\sigma_{\text{loc}}^2(K, T)K^2 \partial_{KK}^2 C,$$

that is

$$\frac{\partial C}{\partial T} = \frac{1}{2}\sigma_{\text{loc}}^2(K, T)K^2 \frac{\partial^2 C}{\partial K^2} - (r-q)K \frac{\partial C}{\partial K} - qC, \quad (16)$$

which is (2) and rearranges to (11). Conversely, given prices satisfying (10), define σ_{loc} by (11); it is well defined and non-negative precisely because of (10), and the computation above run backwards shows that the corresponding diffusion has marginals with the prescribed call prices. Uniqueness is immediate: (16) determines $\sigma_{\text{loc}}^2(K, T)$ pointwise from the price surface. $\square$

Remark 4.2 (Sign conventions). A frequent source of error is the sign of the qC term. The statement in (2), with $+qC$ on the right-hand side, and the equivalent form (11) with $+qC$ in the numerator, are consistent: moving qC across in (16) changes its sign. With $q = 0$ the familiar textbook form $\partial_T C = \frac{1}{2}\sigma_{\text{loc}}^2 K^2 \partial_{KK}^2 C - rK \partial_K C$ is recovered.

4.2 What Dupire's theorem does not say

Theorem 4.1 is a statement about *marginal distributions*. It says that among diffusions of the form (1) there is exactly one matching the marginals $\{\mathcal{L}(S_T)\}_{T>0}$. It does not say that the market's underlying follows that diffusion, and the difference is not academic.

Theorem 4.3 (Gyöngy, 1986). *Let $dX_t = \beta_t dt + \delta_t dW_t$ be an Itô process with bounded $\delta \delta^\top$. Define*

$$\bar{b}(x, t) = \mathbb{E}[\beta_t | X_t = x], \quad \bar{a}(x, t) = \mathbb{E}[\delta_t \delta_t^\top | X_t = x].$$

Then the SDE $dY_t = \bar{b}(Y_t, t)dt + \bar{a}(Y_t, t)^{1/2}dW_t$ admits a weak solution with $\mathcal{L}(Y_t) = \mathcal{L}(X_t)$ for every t .

Applied to a stochastic-volatility model $dS_t = (r-q)S_t dt + \sqrt{v_t}S_t dW_t$, Theorem 4.3 identifies the local volatility as a conditional expectation,

$$\sigma_{\text{loc}}^2(K, T) = \mathbb{E}^{\mathbb{Q}}[v_T | S_T = K]. \quad (17)$$

Local volatility is the projection of the true instantaneous variance onto the terminal spot. Projections lose information. Two models with the same projection — hence the same vanilla prices, by Theorem 4.1 — can have entirely different forward smiles, different spot/volatility covariance dynamics, and therefore different prices for any payoff that is not a function of a single marginal. Section 8.6 measures this gap.

4.3 Dupire in implied-variance coordinates

Equation (11) is numerically hostile: the denominator is a small difference of large numbers, and C ranges over several orders of magnitude across the surface. Gatheral's reformulation is far better behaved because $w = O(10^{-2})$ and varies smoothly.

Theorem 4.4 (Gatheral, 2006, eq. (1.10)). *With $y = \log(K/F_T)$ and $w(y, T) = \sigma_{\text{BS}}^2 T$, Dupire's local variance is*

$$\sigma_{\text{loc}}^2(y, T) = \frac{\partial_T w(y, T)}{g(y, T)}, \quad (18)$$

where the derivative $\partial_T w$ is taken at fixed y , and g is Durrleman's function (8).

Proof sketch. Substitute $C = C^{\text{BS}}(y, w(y, T))$ into (16). The numerator of (11) becomes, after using $\partial_T C = \partial_w C^{\text{BS}} \cdot \partial_T w$ and $\partial_w C^{\text{BS}} = \frac{1}{2} e^{-rT} F_T \varphi(d_+) / \sqrt{w}$, a positive multiple of $\partial_T w$; the denominator, by the computation in Proposition 3.4, is the same positive multiple times $g(y, T)$. The common factor $\frac{1}{2} e^{-rT} F_T \varphi(d_+) / \sqrt{w}$ cancels, leaving (18). The full computation is in Appendix B. $\square$

Corollary 4.5 (Well-posedness of the extraction). *The local variance (18) is finite and strictly positive on a region of the surface if and only if, on that region, the slice is free of butterfly arbitrage ($g > 0$) and the surface is free of calendar-spread arbitrage ($\partial_T w > 0$).*

This corollary is the organising principle of the report. It says that the two static-arbitrage conditions are *exactly* the numerator and denominator conditions of Dupire's formula. A surface that admits butterfly arbitrage does not merely produce a slightly wrong local volatility; it produces an infinite or imaginary one. Fitting an arbitrage-free surface is therefore not a matter of financial hygiene. It is the regularisation.

5 Ill-posedness of the naive extraction

5.1 Differentiation as an inverse problem

By Theorem 3.1, extracting the risk-neutral density from prices is the application of ∂_{KK}^2 . On L^2 , differentiation is the inverse of the compact integration operator and is therefore unbounded: arbitrarily small perturbations of the data produce arbitrarily large perturbations of the output. Concretely, adding $\varepsilon \sin(\omega K)$ to C perturbs $\partial_{KK}^2 C$ by $-\varepsilon \omega^2 \sin(\omega K)$: high-frequency noise is amplified by ω^2 . This is the textbook example of a mildly ill-posed problem in the sense of Hadamard (Tikhonov and Arsenin, 1977): a solution exists and is unique, but does not depend continuously on the data.

5.2 An exact error budget for the second difference

The following makes the amplification quantitative for the estimator one actually uses.

Proposition 5.1. *Let $\hat{C}(K) = C(K) + \varepsilon_K$ with ε_K i.i.d., mean zero, standard deviation ε , and let $C \in C^4$. Define the three-point second difference*

$$\hat{D}_h = \frac{\hat{C}(K+h) - 2\hat{C}(K) + \hat{C}(K-h)}{h^2}.$$

Then

$$\mathbb{E}[\hat{D}_h] - C''(K) = \frac{h^2}{12} C^{(4)}(K) + O(h^4), \quad \text{sd}(\hat{D}_h) = \frac{\sqrt{6}\varepsilon}{h^2}, \quad (19)$$

so that, to leading order,

$$\text{RMSE}(h) \approx ah^2 + \frac{b}{h^2}, \quad a = \frac{|C^{(4)}(K)|}{12}, \quad b = \sqrt{6}\varepsilon. \quad (20)$$

The minimiser and minimal value are

$$h^* = \left(\frac{b}{a}\right)^{1/4} = \left(\frac{12\sqrt{6}\varepsilon}{|C^{(4)}|}\right)^{1/4}, \quad \text{RMSE}(h^*) = 2\sqrt{ab} = \sqrt{\frac{\sqrt{6}|C^{(4)}|\varepsilon}{3}}. \quad (21)$$

Proof. Taylor expansion of C at K to fourth order gives $\frac{C(K+h) - 2C(K) + C(K-h)}{h^2} = C'' + \frac{h^2}{12} C^{(4)} + O(h^4)$, the odd orders cancelling by symmetry. For the variance, the noise enters as $(\varepsilon_{K+h} - 2\varepsilon_K + \varepsilon_{K-h})/h^2$, whose variance is $(1 + 4 + 1)\varepsilon^2/h^4 = 6\varepsilon^2/h^4$. Adding bias squared and variance and taking the square root gives (20). Setting $\frac{d}{dh}(ah^2 + bh^{-2}) = 2ah - 2bh^{-3} = 0$ yields $h^4 = b/a$ and, substituting, $\text{RMSE} = a\sqrt{b/a} + b\sqrt{a/b} = 2\sqrt{ab}$. $\square$

Three consequences deserve emphasis.

1. **The $\varepsilon^{1/2}$ barrier.** By (21) the best achievable accuracy scales as $\varepsilon^{1/2}$. Halving the bid–ask spread buys a factor $\sqrt{2} \approx 1.41$, not a factor 2. There is no amount of tick-size improvement that makes naive differentiation competitive.
2. **The optimal step is not available.** h^* grows like $\varepsilon^{1/4}$ and, for realistic ε and the convexity of index options, exceeds the listed strike spacing — see Section 8, where $h^* \approx 10$ strike points against a ladder spaced at 2.5. One cannot choose the optimal stencil because the data are not there. Any admissible h sits on the steep noise-dominated branch.
3. **The problem compounds.** Dupire’s formula divides by this quantity. A relative error of 30% in $\partial_{KK}^2 C$ is a relative error of 30% in σ_{loc}^2 at best, and a sign change — hence a meaningless answer — whenever the noise exceeds the signal, which happens systematically in the wings where $\partial_{KK}^2 C$ is small.

5.3 Regularisation

The standard remedies all replace the data by a smooth surrogate, and differ in how the surrogate is constrained.

- **Tikhonov / smoothing splines.** Fengler (2009) fits a natural cubic spline to call prices subject to explicit convexity and monotonicity constraints, with a roughness penalty $\lambda \int (C'')^2$. Non-parametric and flexible; requires choosing λ and does not by itself control the maturity direction.
- **Discrete arbitrage-free interpolation.** Andreasen and Huge (2011) solve a one-step implied finite-difference problem, producing prices that are arbitrage-free by construction on the discrete grid.
- **Parametric projection.** Fit a low-dimensional family whose members are arbitrage-free by construction. This is the route taken here. The regularisation is implicit: the model class *is* the penalty, and the number of parameters (3 per maturity) is far below the number of quotes, so the fit necessarily smooths.

We note a pleasant empirical consequence, documented in Section 8: because the fit averages over many noisy quotes, the fitted surface is *closer to the truth than the data it was fitted to*. Regularisation here does not trade accuracy for stability; it buys both.

6 Arbitrage-free parameterisation

6.1 Raw SVI

Gatheral (2004) proposed the five-parameter *raw SVI* slice

$$w(y) = a + b \left(\rho(y - m) + \sqrt{(y - m)^2 + \varsigma^2} \right), \quad (22)$$

with $b \geq 0$, $|\rho| < 1$, $\varsigma > 0$ and $a + b\varsigma\sqrt{1 - \rho^2} \geq 0$ (which ensures $w \geq 0$). Its asymptotic slopes are $b(1 \mp \rho)$ as $y \rightarrow \mp\infty$, so Lee (2004)’s moment formula forces

$$b(1 + |\rho|) \leq \frac{4}{T}, \quad (23)$$

a necessary condition for the absence of butterfly arbitrage. SVI fits equity smiles remarkably well — it is exact in the large-maturity Heston limit (Gatheral, 2011) — but says nothing about how neighbouring slices relate, so a slice-by-slice SVI fit routinely produces calendar arbitrage.

6.2 SSVI and eSSVI

[Gatheral and Jacquier \(2014\)](#) tie the slices together through the ATM total variance $\theta_T = w(0, T)$:

$$w(y, \theta) = \frac{\theta}{2} \left\{ 1 + \rho \varphi(\theta)y + \sqrt{(\varphi(\theta)y + \rho)^2 + 1 - \rho^2} \right\}. \quad (24)$$

This is the only known low-dimensional family with *closed-form* sufficient conditions for the absence of static arbitrage on the whole surface. We use the extension of [Hendriks and Martini \(2019\)](#), *eSSVI*, in which $\varphi(\theta)$ is replaced by a free $p_i > 0$ for each quoted maturity. Each slice is then a three-parameter object (θ_i, ρ_i, p_i) , and is a subfamily of raw SVI (22) with

$$a = \frac{\theta}{2}(1 - \rho^2), \quad b = \frac{\theta p}{2}, \quad m = -\frac{\rho}{p}, \quad \varsigma = \frac{\sqrt{1 - \rho^2}}{p}. \quad (25)$$

Two features make it the right choice here. First, $w(0) = \theta$ exactly, so the ATM term structure is *bootstrapped* rather than fitted, and the level of the surface is never traded off against its shape. Second, the butterfly conditions of [Gatheral and Jacquier \(2014, Thm. 4.2\)](#) specialise to the two closed-form inequalities

$$\theta p(1 + |\rho|) < 4, \quad \theta p^2(1 + |\rho|) \leq 4. \quad (26)$$

From (25), the first is exactly Lee's bound (23) written in eSSVI parameters. We impose (26) as hard constraints in the optimiser and verify $g > 0$ numerically on a dense grid afterwards.

6.3 Calendar-spread arbitrage and interpolation in maturity

Condition (C) has two distinct parts, and conflating them is the commonest error in implementations. Between two quoted maturities we must have $w_{i+1}(y) \geq w_i(y)$ for all y , which we impose as an explicit inequality constraint on a grid of 281 log-moneyness points during a *sequential*, short-maturity-first calibration. But we must also interpolate to maturities that are not quoted, and here a natural-looking choice fails.

Remark 6.1 (A failure we encountered). Interpolating the slice *parameters* (θ, ρ, p) across maturities with a monotone spline does *not* preserve calendar monotonicity of w , even when every node slice satisfies $w_{i+1} \geq w_i$. In our data this produced $\min_{y,T} \partial_T w = -7.5 \times 10^{-3}$: genuine calendar arbitrage created purely by the interpolation, and with it a negative local variance. The fix is to interpolate the total variance itself.

Lemma 6.2 (Structural exclusion of calendar arbitrage). *Fix y and suppose $w_1(y) \leq w_2(y) \leq \dots \leq w_n(y)$. Let $T \mapsto \hat{w}(y, T)$ be the PCHIP (piecewise cubic Hermite, Fritsch–Carlson) interpolant through $\{(T_i, w_i(y))\}_{i=1}^n$. Then $\partial_T \hat{w}(y, T) \geq 0$ on $[T_1, T_n]$.*

Proof. On each interval $[T_i, T_{i+1}]$ let $\Delta_i = (w_{i+1} - w_i)/(T_{i+1} - T_i) \geq 0$ be the secant slope. The PCHIP rule ([Fritsch and Carlson, 1980](#)) sets the nodal derivative d_i to zero whenever $\Delta_{i-1}\Delta_i \leq 0$, and otherwise to the weighted harmonic mean of Δ_{i-1} and Δ_i , which lies between them and in particular has the same sign; hence $d_i \geq 0$ for all i . A cubic Hermite piece with data $(w_i, w_{i+1}, d_i, d_{i+1})$ is monotone non-decreasing provided $(d_i/\Delta_i, d_{i+1}/\Delta_i)$ lies in the Fritsch–Carlson region, and the harmonic-mean choice guarantees $d_i \leq 3\Delta_i$ and $d_{i+1} \leq 3\Delta_i$, which is a sufficient condition. Therefore each piece is non-decreasing, and so is the interpolant. $\square$

Outside the quoted range we extrapolate at constant implied volatility, $\hat{w}(y, T) = w_1(y)T/T_1$ for $T < T_1$ and $\hat{w}(y, T) = w_n(y)T/T_n$ for $T > T_n$, which is again increasing in T and continuous at the endpoints. By Lemma 6.2 the numerator of (18) is non-negative everywhere by construction; only the denominator has to be audited.

6.4 The calibration algorithm

Algorithm 1: Sequential arbitrage-constrained eSSVI calibration

Input: quotes (K_j, T_j, mid_j) , spot S_0 , rates r, q

Output: arbitrage-free total-variance surface $w(y, T)$

Convert each quote to an implied volatility by Proposition 2.1; discard quotes violating (6) or below the liquidity floor;

for $i = 1, \dots, n$ (*maturities in increasing order*) **do**

Seed $\theta_i^{(0)}$ by interpolating the observed ATM total variance;

Solve, by SLSQP with vega weights ω_j ,

$$\min_{\theta, \rho, p} \sum_{j: T_j = T_i} \omega_j \left(\sqrt{w(y_j; \theta, \rho, p) / T_i} - \sigma_{\text{BS}}^{\text{mid}}(y_j) \right)^2$$

subject to the butterfly conditions (26) and, for $i > 1$, to $w(y; \theta, \rho, p) \geq w_{i-1}(y)$ on a grid of 281 points in $[-1.4, 1.4]$;

Restart from 16 initial points (ρ_0, p_0) and keep the feasible minimiser;

end

Define $w(y, T)$ by PCHIP interpolation of $\{w_i(y)\}$ in T (Lemma 6.2);

Audit $g(y, T) > 0$ and $\partial_T w(y, T) \geq 0$ on a dense grid; check $\int p(y, T) dy = 1$;

Weighting by Black–Scholes vega is not cosmetic. By Remark 2.2, an error of one tick corresponds to a volatility error of $1/\mathcal{V}$; weighting the squared volatility residuals by $\mathcal{V}$ is (up to a constant) weighting the squared *price* residuals, which is the homoscedastic objective.

7 Numerical implementation

7.1 The data-generating process and its pricer

The hidden process is Heston (Heston, 1993),

$$dS_t = (r - q)S_t dt + \sqrt{v_t} S_t dW_t^1, \quad dv_t = \kappa(\theta - v_t)dt + \xi \sqrt{v_t} dW_t^2, \quad d\langle W^1, W^2 \rangle_t = \rho dt, \quad (27)$$

with $S_0 = 100$, $r = 2\%$, $q = 1\%$, and $(v_0, \kappa, \theta, \xi, \rho) = (0.0348, 1.15, 0.0560, 0.55, -0.72)$. The Feller ratio $2\kappa\theta/\xi^2 = 0.43 < 1$ is deliberately violated, as it typically is on calibrated equity surfaces; this is what makes a naive Euler scheme for the variance unusable and motivates the QE scheme below.

Prices are computed by the COS method of Fang and Oosterlee (2008). The risk-neutral density of $y_T = \log(S_T/K)$ is expanded on a cosine basis over a truncated interval $[a, b]$ determined by the cumulants of $\log(S_T/S_0)$,

$$C = K e^{-rT} \sum_{k=0}^{N-1} {}' \text{Re} \left\{ \phi\left(\frac{k\pi}{b-a}\right) e^{ik\pi \frac{x-a}{b-a}} \right\} U_k,$$

with analytic payoff coefficients U_k and the prime denoting a halved first term. We use the “little Heston trap” form of the characteristic function (Albrecher et al., 2007), which keeps the complex logarithm on its principal branch, and we price *puts*, recovering calls by parity, because the put payoff coefficients avoid the cancellation that afflicts the call expansion at long maturities. With $N = 2^{15}$ and a truncation width of $L = 20$ standard deviations, the implementation agrees with an independent Lewis-type contour integral to 10^{-8} absolute (Table 1).

Table 1: Validation of the COS implementation against an independent Lewis-type contour integral, over $T \in \{0.05, 0.5, 2\}$ and $K \in \{60, 100, 160\}$ with the parameters of Section 8. N is the number of cosine terms and L the width of the truncation range in standard deviations. Put–call parity is satisfied to 7.1×10^{-15} at every setting. The production runs use $N = 2^{15}$, $L = 20$.

N	L	max absolute price error
2^{13}	12	1.5×10^0
2^{14}	16	5.0×10^{-4}
2^{15}	20	1.0×10^{-8}
2^{16}	24	6.0×10^{-12}

The failure of the coarsest setting is instructive rather than accidental: at $T = 0.05$ the cumulant-based truncation range is narrow, and a strike at $K = 160$ falls outside it, so the density is truncated where the payoff is still active. A short maturity combined with a distant strike is precisely the regime in which Fourier methods must be used with care, and it is also the regime in which the quotes themselves are least informative.

7.2 Synthetic market design

Exact prices are degraded by three mechanisms, chosen because they are the three that actually destroy the differentiation step:

1. a vega-proportional quoting error, $\text{price} \mapsto \text{price} + \mathcal{V}\eta$ with $\eta \sim \mathcal{N}(0, 40\text{bp}^2)$, i.e. a flat error in volatility units;
2. rounding to a one-cent tick, which is a 5% price error on a twenty-cent option and much worse in volatility terms;
3. a bid–ask spread of $\max(1.5\% \times \text{mid}, 2 \text{ cents})$ on each side.

Only out-of-the-money contracts are quoted (puts below the forward, calls above), on a ladder spaced at 2.5 points near the money and 5 points in the wings, restricted to $|y| \leq 3.2\sqrt{0.05T}$. This reproduces the characteristic asymmetry of listed index markets, in which the downside is quoted much further out than the upside. Quotes below five cents are dropped as illiquid.

7.3 Tabulation of the local-volatility surface

For simulation we need $\sigma_{\text{loc}}(S, t)$ at arbitrary (S, t) , whereas (18) gives it in (y, T) coordinates with $y = \log(K/F_T)$. Since the strike argument of Dupire’s formula becomes the spot argument of the SDE, we evaluate at $y = \log S_t - \log S_0 - (r - q)t$. The surface is tabulated on a 221×121 grid in (y, t) and interpolated bilinearly — deliberately not by splines, which overshoot near the wing boundary. Outside the quoted region the local volatility is frozen at the boundary value. This is the standard desk convention and it has a mathematical justification: Dupire’s equation carries *no information* where there are no quotes, and flat extrapolation keeps the diffusion coefficient bounded, which the Euler scheme requires for weak convergence.

7.4 Monte Carlo

We simulate the log-spot,

$$X_{t_{n+1}} = X_{t_n} + \left(r - q - \frac{1}{2}\sigma_{\text{loc}}^2(e^{X_{t_n}}, t_n)\right)\Delta + \sigma_{\text{loc}}(e^{X_{t_n}}, t_n)\sqrt{\Delta}Z_{n+1}, \quad (28)$$

which keeps the spot positive by construction and has weak order 1 for bounded, Lipschitz coefficients (Kloeden and Platen, 1992). Variance reduction uses antithetic paths together with the martingale control variate $S_T - F_T$, whose expectation is known exactly; the optimal loading is estimated by regression.

Barriers require care. Discrete monitoring of a continuously-monitored barrier biases knock-out prices upward by $O(\sqrt{\Delta})$ (Broadie et al., 1997), an error of several percent at daily monitoring. We use the Brownian-bridge correction: conditionally on the endpoints of a step, X is a Brownian bridge whose running maximum satisfies

$$\mathbb{P}\left(\max_{t \in [t_n, t_{n+1}]} X_t \leq \log B \mid X_{t_n} = x, X_{t_{n+1}} = x'\right) = 1 - \exp\left(-\frac{2(\log B - x)(\log B - x')}{\sigma^2 \Delta}\right) \quad (29)$$

for $x, x' < \log B$. Sampling a uniform against (29) at each step restores $O(\Delta)$ bias.

The Heston reference engine uses the Quadratic-Exponential scheme of Andersen (2008), which matches the first two conditional moments of the CIR variance by a switched moment-matching rule (a shifted squared Gaussian when the conditional coefficient of variation is small, an exponential with an atom at zero when it is large), together with the martingale correction for the log-spot. This is essential here because the Feller condition is violated.

8 Results

All figures and tables below are produced by the companion notebook (Appendix C) from a single seeded run; total runtime is about four minutes. The data-generating process is Heston with $v_0 = 0.0348$, $\kappa = 1.15$, $\theta = 0.0560$, $\xi = 0.55$, $\rho = -0.72$, on a spot $S_0 = 100$ with $r = 2\%$, $q = 1\%$. The Feller ratio $2\kappa\theta/\xi^2 = 0.426$ is well below one, so the variance process touches zero: this is a deliberately unfriendly test case.

8.1 The market and the inversion

The quote file contains 242 out-of-the-money contracts across eight maturities from one month to two years. After tick rounding and the addition of vega-proportional quoting error, the realised mismatch between the observable mid-quote and the true implied volatility has root-mean-square 39.5 bp and maximum 100.8 bp. No quote violates the static bounds of Proposition 2.1, so the inversion never fails; 66 quotes are nevertheless discarded by a liquidity filter (mid below five cents), leaving 176. That the discarded contracts are all deep out-of-the-money is the practical face of Remark 2.2: where vega is small, a one-tick price error is a very large volatility error.

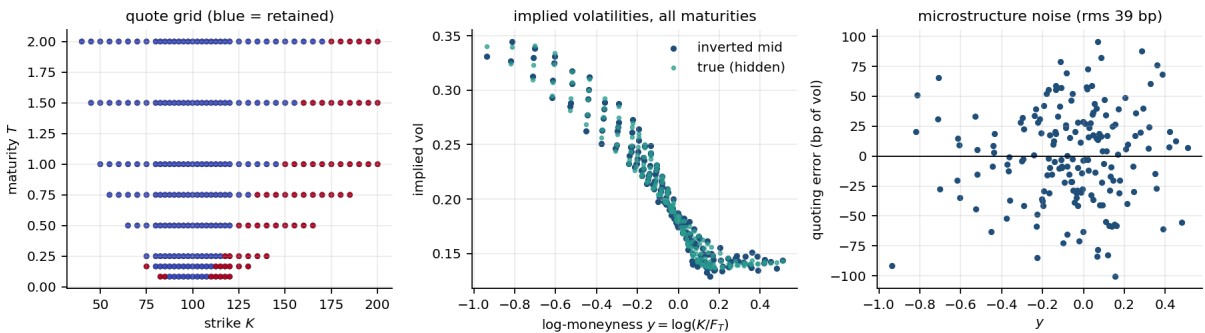


Figure 1: The synthetic market. Left: the quote grid, with retained contracts in blue. Centre: inverted mid volatilities against the true (unobservable) values. Right: the realised quoting error in volatility terms, largest in the wings where vega is smallest.

8.2 Calibration and the arbitrage audit

Table 2 reports the bootstrapped eSSVI parameters. The ATM total variance θ_i is increasing by construction; the slice correlation ρ_i is stable around -0.65 , and the curvature parameter p_i decays with maturity, which is the familiar flattening of the smile in log-moneyness as T grows.

Table 2: Bootstrapped eSSVI parameters. The last column is the weighted root-mean-square fit residual of the slice, in volatility units.

T (y)	quotes	θ_i	ρ_i	p_i	slice rmse
0.0833	9	0.00283	-0.6346	8.050	2.71×10^{-3}
0.1667	13	0.00535	-0.7808	6.991	2.69×10^{-3}
0.2500	16	0.00814	-0.7051	6.860	4.11×10^{-3}
0.5000	20	0.01554	-0.6667	6.411	4.79×10^{-3}
0.7500	24	0.02430	-0.6870	5.126	4.23×10^{-3}
1.0000	28	0.03186	-0.6489	5.071	3.86×10^{-3}
1.5000	31	0.04971	-0.6491	3.991	2.95×10^{-3}
2.0000	35	0.06797	-0.6419	3.327	4.10×10^{-3}

The single most informative number in this report is the pair

$$\text{RMSE}(\hat{\sigma}, \sigma^{\text{mid}}) = 41.8 \text{ bp}, \quad \text{RMSE}(\hat{\sigma}, \sigma^{\text{true}}) = 21.3 \text{ bp}.$$

The fitted surface is *twice as close to the truth as the data it was fitted to*. This is not a coincidence or an artefact of a favourable draw: it is exactly what a well-chosen regularisation is supposed to do. The eSSVI family has 24 parameters for 176 quotes and is smooth in both arguments, so independent quoting errors are averaged out; the residual 21.3 bp is the model error of the family itself, not noise. Fitting an interpolant with enough freedom to honour every mid exactly would have driven the first number to zero and the second one up.

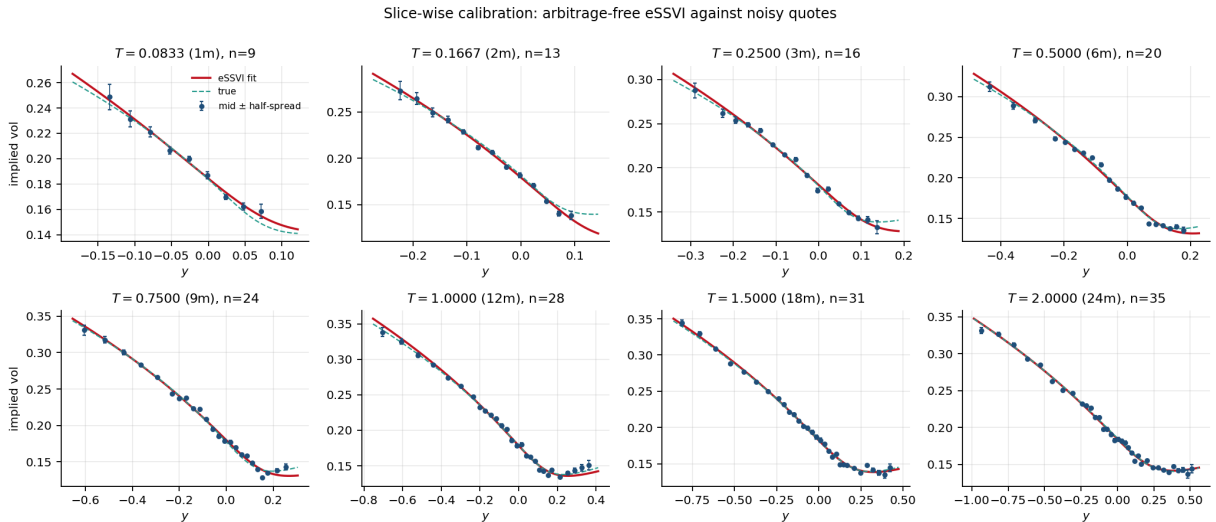


Figure 2: Slice-by-slice calibration. Error bars are half the bid–ask spread. The fitted curve (red) tracks the true smile (dashed green) more closely than the individual mid-quotes do, most visibly in the short-dated wings.

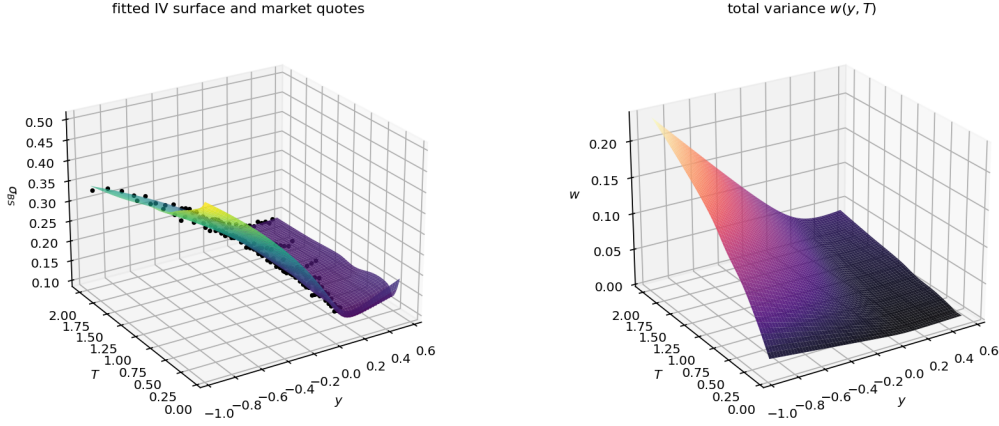


Figure 3: The calibrated implied-volatility surface (left) with the market quotes overlaid, and the corresponding total-variance surface $w(y, T)$ (right). The monotone increase of w in T visible on the right panel is the calendar-spread condition.

The static-arbitrage audit, run on a 221×160 grid covering $|y| \leq 1.1$ and $T \in [T_1, T_n]$, returns

$$\min_{y,T} g(y, T) = 0.3007 > 0, \quad \min_{y,T} \partial_T w(y, T) = 8.63 \times 10^{-5} > 0,$$

and the implied density integrates to 1.000000 on every slice to six decimal places. Both conditions hold with a comfortable margin rather than marginally, which matters: by Corollary 4.5, g is the denominator of Dupire’s formula, so a surface that is only barely butterfly-free produces a local volatility that is barely finite.

It is worth recording a failure here, because it was instructive. Our first implementation interpolated the *slice parameters* (θ_i, ρ_i, p_i) in maturity with a monotone cubic. Every calibrated slice satisfied the closed-form conditions (26), yet the audit returned $\min \partial_T w = -7.5 \times 10^{-3}$: the interpolated surface contained calendar arbitrage between the nodes. Slice-wise arbitrage-freeness does not survive interpolation of the parameters. Lemma 6.2 — interpolating w itself at fixed y — is what removes the problem structurally.

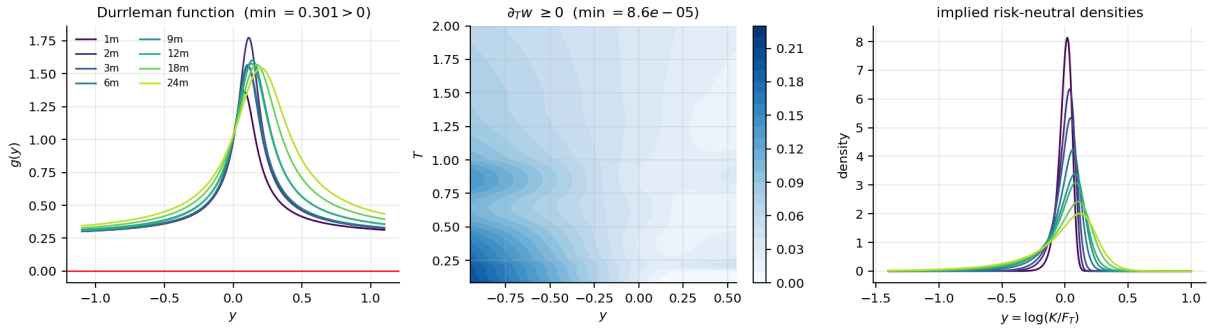


Figure 4: Arbitrage diagnostics. Left: Durrleman’s function, bounded away from zero on every slice. Centre: $\partial_T w \geq 0$ over the whole domain. Right: the implied risk-neutral densities, each non-negative and of unit mass.

8.3 The ill-posedness, measured

Proposition 5.1 is confirmed quantitatively. At $K = 100$, $T = 1$ the exact model values are $\partial_{KK}^2 C = 2.0968 \times 10^{-2}$ and $C^{(4)} = 1.259 \times 10^{-5}$. With a quoting noise of one cent, (21) predicts

$$h^* = \left(\frac{12\sqrt{6}\varepsilon}{|C^{(4)}|} \right)^{1/4} = 12.36,$$

against a listed strike spacing of 2.5. The optimal finite-difference step is *five times wider than the strikes actually quoted*, and at the available spacing the noise term dominates the truncation term by more than an

order of magnitude. The Monte-Carlo experiment in the left panel of Figure 5, averaging 400 noise draws at each step size, reproduces the predicted $ah^2 + bh^{-2}$ curve including the location of its minimum. The right panel shows the consequence: the best attainable accuracy falls only as $\varepsilon^{1/2}$, so halving the bid–ask spread buys a factor $\sqrt{2}$.

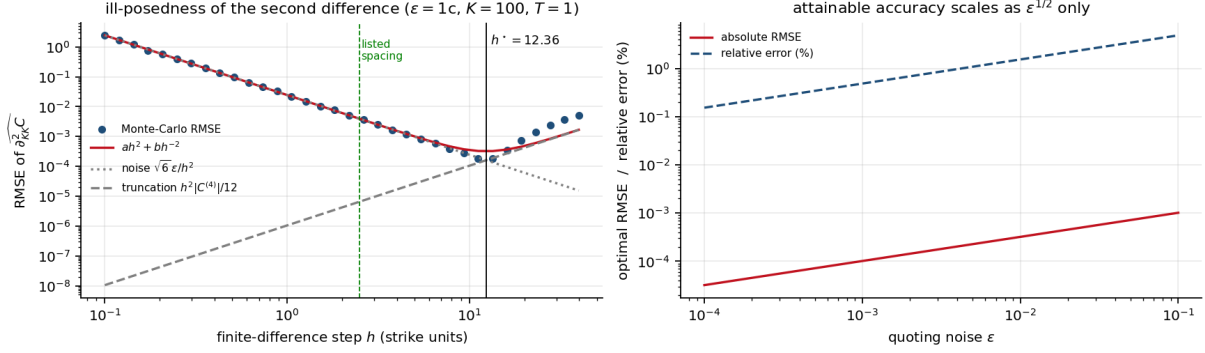


Figure 5: The second difference as an ill-posed problem. Left: measured RMSE against step size, with the two competing error terms. Right: the optimal accuracy as a function of the noise level.

Applying Dupire’s formula directly to the quote ladder, as one is tempted to do, fails exactly as predicted. Of the 160 nodes at which a second difference in strike and a difference in maturity can both be formed, 42 — **26.2%** — return an inadmissible local variance: negative, or corresponding to a volatility above 200%. The negative values are not a numerical curiosity; they are the statement that the estimated $\partial_{KK}^2 C$ is negative, i.e. that the noisy quote ladder is not convex and the implied “density” is not a density. On the surviving nodes the estimator still differs from the regularised one by 18.1 volatility points in root-mean-square.

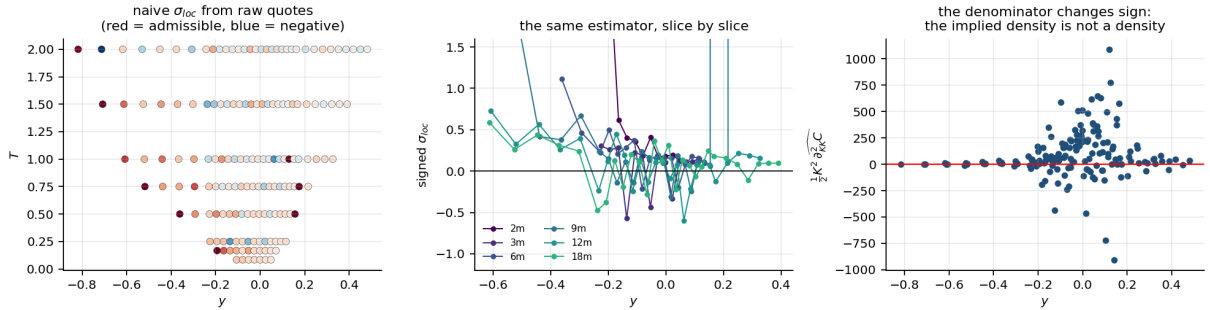


Figure 6: Dupire’s formula applied to raw quotes. Left: admissible (red) and negative (blue) nodes. Centre: the same estimator slice by slice, oscillating by more than 100 volatility points between adjacent strikes. Right: the denominator changes sign.

8.4 The local-volatility surface

Two validations precede the result itself.

Internal consistency. Dupire’s price form (11) and Gatheral’s implied-variance form (18) are the same theorem in different coordinates, so applying both to the *same* calibrated surface must give the same answer; any difference is an implementation error. Table 3 shows agreement to between 1.6×10^{-5} and 3.4×10^{-4} in relative terms, the residual being the finite-difference error of the price form (largest near the maturity nodes, where the PCHIP interpolant is only C^1 and $\partial_T w$ is merely Lipschitz).

Table 3: Cross-validation of the two Dupire forms on the calibrated surface, over strikes $K \in \{70, 85, 95, 100, 110, 125\}$.

T	max abs. difference in σ_{loc}	max relative difference
0.30	4.84×10^{-5}	3.44×10^{-4}
0.62	9.51×10^{-6}	7.93×10^{-5}
0.88	3.04×10^{-5}	1.23×10^{-4}
1.25	4.44×10^{-6}	2.37×10^{-5}
1.75	2.10×10^{-6}	1.55×10^{-5}

External validation. Because the data-generating process is known, the true local-volatility surface can be computed by applying (11) to Heston prices evaluated to machine accuracy, where finite differencing is harmless. Against that benchmark:

region	rmse (vol pts)	max (vol pts)
core, $ y < 0.7 \times \text{quoted band}$	0.98	6.87
full quoted band	1.38	10.77
extrapolated, outside the band	5.90	37.45

Inside the region where options actually trade the surface is recovered to about one volatility point; outside it the error grows by a factor of six. This is not a defect of the method. Dupire’s equation carries no information where there are no quotes, and any value reported there is a property of the extrapolation convention, not of the market. A desk that prices a barrier struck outside the quoted band is pricing its own extrapolation.

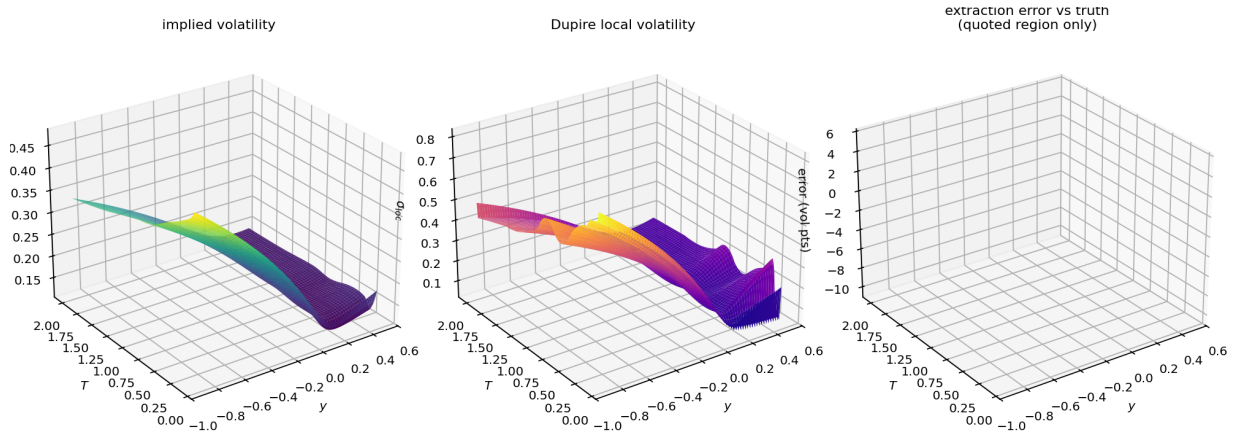


Figure 7: Implied volatility, the extracted local volatility, and the extraction error against ground truth inside the quoted region. The local-volatility surface is markedly steeper in log-moneyness than the implied surface.

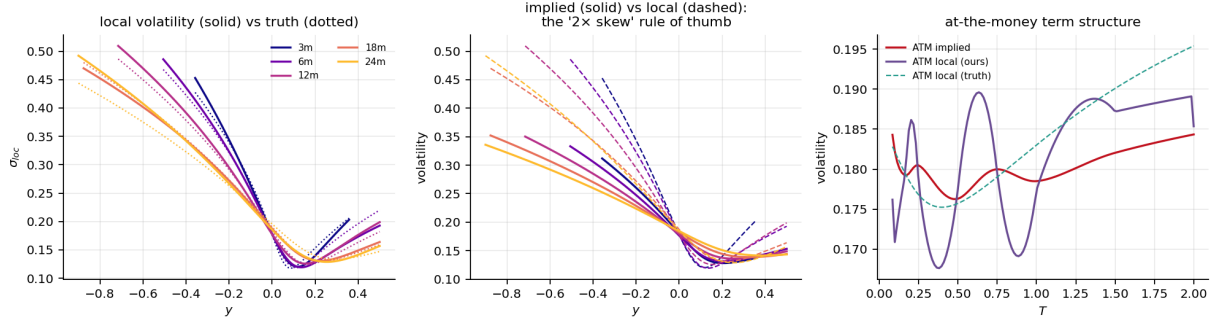


Figure 8: Left: extracted local volatility (solid) against truth (dotted). Centre: implied (solid) and local (dashed) volatility on the same axes, illustrating the rule of thumb that local volatility has roughly twice the skew of implied volatility. Right: the at-the-money term structures.

The centre panel of Figure 8 shows the classical “factor of two” rule: near the money the local-volatility skew is approximately twice the implied-volatility skew. This follows from (18) by expanding to first order in y and is a useful sanity check on any implementation.

8.5 Dupire consistency and Monte-Carlo convergence

The decisive test is to simulate the extracted local-volatility SDE and reprice the very options the surface was built from. Theorem 4.1 says the implied volatilities must come back. Over 24 contracts spanning four maturities and the full quoted moneyness range, the root-mean-square discrepancy is

$$\text{rms}|\sigma^{\text{MC}} - \sigma^{\text{surface}}| = 13.0 \text{ bp}, \quad \text{mean Monte-Carlo standard error} = 4.2 \text{ bp}.$$

The residual is structured rather than random: it is -25 to -35 bp at $T = 0.25$ and falls below 5 bp for $T \geq 1$. The cause is the local-volatility table, which is interpolated linearly in (y, t) on a 221×121 grid; at short maturities σ_{loc} varies rapidly and the interpolation slightly understates the curvature. It is an interpolation error, not a violation of Dupire’s theorem, which is confirmed by the fact that refining the time step does not remove it.

Table 4: Weak convergence of the log-Euler scheme, at the money, $T = 1$, against a reference run with 1600 steps and common random numbers. Errors are in basis points of implied volatility; the Monte-Carlo uncertainty is ± 7.3 bp throughout.

steps per year	12	25	50	100	200
bias (bp)	33.9	14.5	2.0	0.3	1.5

The bias falls from 33.9 bp to below the Monte-Carlo noise floor by 50 steps, consistent with the first-order weak convergence of the scheme. Using a fine-grid run of the same scheme as the reference, rather than the surface value, is essential: the surface target carries the interpolation error just discussed, which does not vanish as $\Delta \rightarrow 0$ and would have masked the convergence entirely.

The barrier machinery is validated separately against the closed-form continuously-monitored up-and-out call under flat Black–Scholes, obtained by integrating the joint law of $(X_T, \max_{t \leq T} X_t)$. With the Brownian-bridge correction of (29), the simulated prices at $B \in \{110, 120, 130, 150\}$ match the closed form with z -scores of 0.34, -0.82 , -1.08 and -1.29 : within Monte-Carlo error at every barrier, including the near barrier $B = 110$ where the naive discrete estimator would be badly biased.

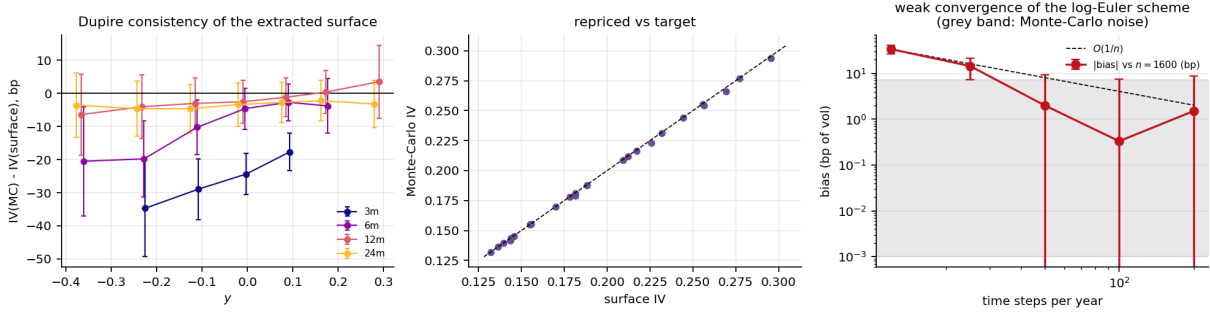


Figure 9: Left: implied-volatility discrepancy between the Monte-Carlo reprice and the calibrated surface, with ± 2 standard errors. Centre: repriced against target. Right: weak convergence of the log-Euler scheme, with the Monte-Carlo noise floor shaded.

8.6 Exotics and model risk

We now price six payoffs under three engines that agree, by construction, on every European option: the calibrated local-volatility model, the true Heston dynamics, and flat Black–Scholes at the one-year at-the-money implied volatility of 17.85%. Any dispersion is model risk in its purest form — it cannot be arbitrated away by recalibrating to vanillas, because all three are already calibrated to vanillas.

Table 5: Exotic prices, $T = 1$, 200,000 paths, 250 steps. Monte-Carlo standard errors in parentheses. Percentages are relative to the true (Heston) price.

	Local vol	Heston (truth)	BS (flat)	LV vs H	BS vs H
European call $K = 100$	7.6074 (0.012)	7.6348 (0.013)	7.5241 (0.012)	−0.4%	−1.4%
Up-and-out call $B = 130$	5.9301 (0.016)	6.2530 (0.017)	3.4827 (0.014)	−5.2%	−44.3%
Asian call $K = 100$	4.5873 (0.012)	4.6302 (0.012)	4.5890 (0.016)	−0.9%	−0.9%
Digital call $K = 110$	0.3367 (0.001)	0.3373 (0.001)	0.2801 (0.001)	−0.2%	−16.9%
Lookback (floating)	15.3181 (0.022)	15.1403 (0.023)	13.2315 (0.027)	+1.2%	−12.6%
Cliquet, cap 8% floor −8%	0.0722 (0.0002)	0.0697 (0.0002)	0.0641 (0.0002)	+3.7%	−8.0%

Three readings of Table 5 deserve comment.

Flat Black–Scholes is not a competitor. Ignoring the smile misprices the digital by 17%, the lookback by 13% and the knock-out by 44%. These errors have the sign one would predict: a flat surface omits the left-tail mass that the skew implies, so path-dependent payoffs sensitive to downside excursions are systematically cheapened, while the knock-out — which *benefits* from the spot not rising — is cheapened because the flat volatility of 17.85% exceeds the local volatility prevailing above the money.

Local volatility is right on marginals and wrong on paths. The European and the digital are essentially exact, as they must be: the digital is $-\partial_K C$, a functional of the terminal marginal alone. Indeed, the decomposition

$$\underbrace{0.2599}_{\text{BS at the smile vol}} + \underbrace{0.0771}_{\text{skew term } -\mathcal{V} \partial_K \sigma} = 0.3370 = \underbrace{-\partial_K C|_{\text{surface}}}_{0.3370} \approx \underbrace{0.3367}_{\text{local-vol MC}}$$

reproduces the local-volatility price to within one Monte-Carlo standard error, confirming that the entire difference from the naive 0.2599 is the skew term. But the knock-out is 5.2% cheap and the cliquet 3.7% rich under local volatility. These are path functionals; by Theorem 4.3 they depend on more than the marginals, and local volatility gets them wrong by construction.

The error grows as the payoff becomes more path-dependent. This is best seen by varying the barrier.

Table 6: Up-and-out call, $K = 100$, $T = 1$, as a function of the barrier level.

B	Local vol	Heston	BS (flat)	LV vs Heston
110	0.2622	0.5935	0.1609	−55.8%
118	2.2579	3.0546	1.0525	−26.1%
125	4.7089	5.2321	2.4166	−10.0%
132	6.2741	6.5061	3.8956	−3.6%
145	7.3051	7.3427	5.9897	−0.5%
165	7.5677	7.6017	7.2095	−0.4%

As $B \rightarrow \infty$ the knock-out becomes a vanilla and the two models must agree; they do, to 0.4%. As the barrier tightens the payoff depends more on the path and less on the terminal value, and the discrepancy grows monotonically to −56% at $B = 110$. The mechanism is the forward smile. Under Heston, volatility is stochastic and can fall after a rise in spot, which lets paths approach the barrier and retreat; under local volatility, volatility is a deterministic function of spot, so a path that has risen *must* carry the low volatility attached to that spot level, and knocks out less erratically. The economic statement is that the vanilla market does not price the conditional law of volatility given spot, and a near knock-out is almost entirely a bet on that conditional law. Reporting a knock-out price from a local-volatility model without a model-risk reserve is, on this evidence, indefensible.

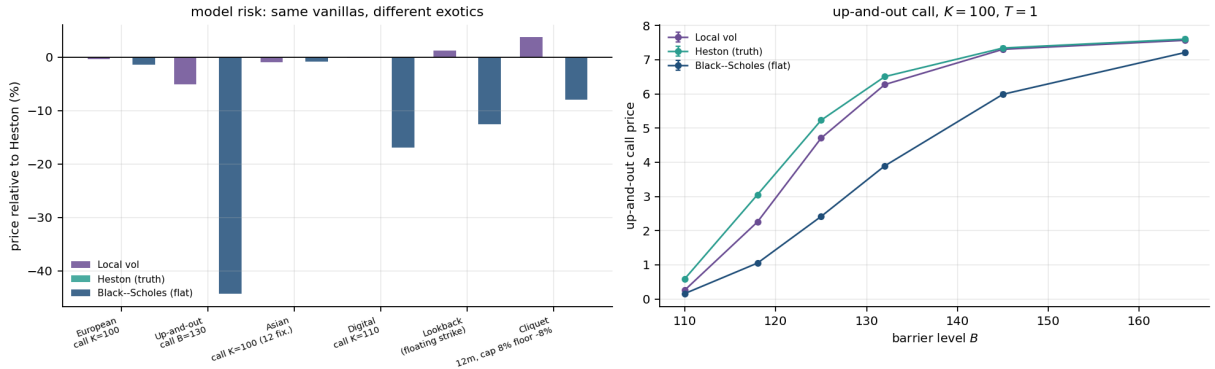


Figure 10: Left: exotic prices relative to the true Heston value. Right: the up-and-out call across barrier levels; the local-volatility and Heston curves separate as the barrier tightens and merge as it recedes.

8.7 How much of the surface is determined by the data?

Finally the error bar. Holding the exact prices fixed and resampling only the microstructure noise, we re-run the entire pipeline 24 times and measure the dispersion of the extracted local volatility. The ratio $\text{sd}(\sigma_{\text{loc}})/\text{sd}(\sigma_{\text{BS}})$ is the amplification factor of the inverse problem after regularisation.

Table 7: Noise amplification (upper block) and absolute dispersion of σ_{loc} in volatility points (lower block), over 24 resamples of the quoting noise.

	$T = 0.25$	$T = 0.50$	$T = 1.00$	$T = 1.50$	$T = 2.00$
<i>amplification</i> $\text{sd}(\sigma_{\text{loc}})/\text{sd}(\sigma_{\text{BS}})$					
$y = -0.50$	5.77	2.64	5.26	3.60	11.04
$y = -0.25$	4.86	3.63	3.71	2.07	9.68
$y = 0.00$	2.83	2.16	3.49	2.03	4.78
$y = +0.15$	3.78	2.01	2.82	2.88	3.76
$y = +0.30$	5.61	1.48	2.86	3.20	5.23
<i>dispersion</i> $\text{sd}(\sigma_{\text{loc}})$, volatility points					
$y = -0.50$	1.85	0.77	0.76	0.63	1.79
$y = -0.25$	0.91	0.53	0.46	0.28	0.90
$y = 0.00$	0.35	0.25	0.37	0.24	0.63
$y = +0.15$	1.15	0.45	0.32	0.22	0.60
$y = +0.30$	2.90	0.79	0.79	0.55	0.74

Even after regularisation the differentiation amplifies noise by a factor of 1.5 to 11. The amplification is smallest at the money and at intermediate maturities, and largest in the wings, at the shortest maturity (where the strike coverage is thinnest) and at the longest (where $\partial_T w$ must be estimated at the edge of the interpolant’s support). In absolute terms the at-the-money local volatility is determined to about a quarter of a volatility point, while the short-dated upside wing carries an uncertainty of nearly three points. A practitioner should read the surface as $\sigma_{\text{loc}} \pm$ these numbers, not as a point estimate; the natural use is to price the exotic on a set of resampled surfaces and report the spread.

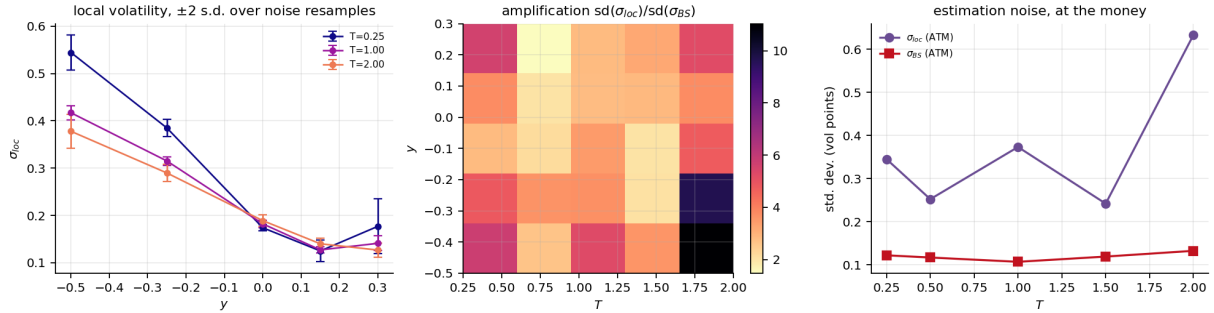


Figure 11: Propagation of quoting noise through the pipeline. Left: local volatility with ± 2 standard deviations over resamples. Centre: the amplification factor. Right: at-the-money dispersion of implied and local volatility against maturity.

9 Discussion

It is worth separating clearly what has been established.

Local volatility is an interpolation device, not a model of dynamics. Theorem 4.1 produces the unique diffusion matching the observed marginals, and the Monte-Carlo test confirms it does so to 13 bp. But Theorem 4.3 says the marginals do not determine the process, and Section 8.6 measures the consequence: up to 56% on a tight knock-out. The correct mental model is that $\sigma_{\text{loc}}(K, T) = \mathbb{E}[v_T | S_T = K]$ is a *conditional average* of the true instantaneous variance, and averaging destroys precisely the information that path-dependent payoffs are sensitive to. Local volatility is therefore excellent as a smile-consistent interpolator — for pricing a vanilla at an unquoted strike, for computing smile-consistent deltas, for

calibrating the marginal leg of a stochastic-local-volatility model — and unreliable as a standalone pricer of path-dependent exotics.

The inverse problem is the whole difficulty. The mathematics of Section 4 is a page of calculus; the engineering of Sections 5–6 is where the project lives. The numbers make this concrete: a 26.2% failure rate for the naive estimator, an optimal step five times the quoted strike spacing, a residual amplification factor of up to 11 even after regularisation. Anyone reporting a local-volatility surface without reporting these diagnostics is reporting a number whose error bar they have not computed.

Regularisation must be chosen, not avoided. There is no neutral option. Cubic splines through the mids, a fine finite-difference grid, an exact interpolant — each is a choice of regulariser, and the ones that fit the data most closely are the ones that behave worst under differentiation. Our choice, projection onto eSSVI, is defensible on three grounds: it is arbitrage-free by construction with closed-form conditions; it has few enough parameters to average out independent quoting errors, which we verified by the 41.8 versus 21.3 bp comparison; and it is smooth, so the derivatives in Dupire’s formula can be computed analytically or by finite differences on a noise-free function. Its cost is model error — the 21.3 bp floor — which a nonparametric method such as Fengler (2009) or Andreasen and Høge (2011) could in principle reduce at the price of weaker smoothness guarantees.

Limitations. The market is synthetic, which is what makes ground-truth validation possible but also means the noise is i.i.d. and the true surface is smooth; real quote errors are cross-sectionally correlated and real surfaces have features (dividend dates, events, pinning) that a three-parameter slice cannot represent. Rates and repo are deterministic and flat. The local-volatility table is interpolated linearly, which we identified as the dominant residual in the short-dated consistency test and which a bicubic scheme with monotonicity constraints would improve. Finally, the natural continuation of the project is a stochastic-local-volatility model: writing $dS_t/S_t = (r - q)dt + \lambda(S_t, t)\sqrt{V_t}dW_t$ and choosing the leverage function λ so that $\lambda^2(K, T)\mathbb{E}[V_T | S_T = K] = \sigma_{\text{loc}}^2(K, T)$ retains the exact vanilla fit established here while restoring the stochastic volatility dynamics whose absence Section 8.6 quantified. The calibration of λ requires a forward Kolmogorov solve or a particle method, and the surface built in this report is exactly its input.

10 Conclusion

We have built an arbitrage-free local-volatility surface from noisy option quotes, end to end, and validated every stage against an independent benchmark. The headline findings are:

1. Implied-volatility inversion is well posed but badly conditioned in the wings, where vega decays exponentially; a one-cent tick moves a deep out-of-the-money implied volatility by a full volatility point.
2. Differentiating quotes directly is hopeless in a precise sense: the RMSE of the second difference is $ah^2 + bh^{-2}$ with $b \propto \varepsilon$, the optimal step (12.4 strike units) is five times the listed spacing (2.5), the best accuracy scales only as $\varepsilon^{1/2}$, and 26.2% of the resulting local variances are negative.
3. Regularisation by an arbitrage-free parameterisation both solves the problem and denoises the data: the fitted surface is twice as close to the truth (21.3 bp) as the quotes it was fitted to (41.8 bp).
4. Arbitrage-freeness is not cosmetic. Durrleman’s function is literally the denominator of Dupire’s formula, so butterfly arbitrage makes the local variance blow up or turn negative. Slice-wise conditions do not survive naive interpolation in maturity; interpolating the total variance itself makes calendar-freeness structural.

5. The extraction is accurate where options trade — about one volatility point of rmse inside the quoted band — and meaningless outside it, where the error is six times larger and reflects only the extrapolation convention.
6. Simulating the extracted surface reproduces the input vanillas to 13 bp, confirming Dupire's theorem numerically; but the same surface misprices a tight knock-out by up to 56% relative to the true dynamics, which is the empirical content of Gyöngy's theorem and the practical limit of what a vanilla market can tell you.

A The Durrleman function from the convexity of the call price

We record the algebra behind Proposition 3.4. Write $F = F_T$, $y = \log(K/F)$, $w = w(y, T)$, and recall that in these coordinates the Black–Scholes call price is

$$C(K, T) = e^{-rT} F \left[\mathcal{N}(d_+) - e^y \mathcal{N}(d_-) \right], \quad d_{\pm} = -\frac{y}{\sqrt{w}} \pm \frac{\sqrt{w}}{2},$$

so that $d_+ - d_- = \sqrt{w}$ and $d_+ d_- = \frac{y^2}{w} - \frac{w}{4}$. Throughout, φ denotes the standard normal density and $' = \partial_y$.

Step 1: first derivative. Since $K = Fe^y$ we have $\partial_K = \frac{1}{K} \partial_y$. Using $\varphi(d_+) = e^y \varphi(d_-)$ — an identity that follows from $d_+^2 - d_-^2 = (d_+ + d_-)(d_+ - d_-) = -2y$ — the terms involving $\partial_y d_+$ and $\partial_y d_-$ collapse and

$$\partial_y C = -e^{-rT} F e^y \left[\mathcal{N}(d_-) + \varphi(d_-) \partial_y d_- \right] \cdot (-1)^0, \quad \partial_y d_- = -\frac{1}{\sqrt{w}} - \frac{w'}{2w} \left(-\frac{y}{\sqrt{w}} \right) - \frac{w'}{4\sqrt{w}}.$$

Collecting,

$$\partial_K C = -e^{-rT} \left[\mathcal{N}(d_-) + \varphi(d_-) \left(\frac{1}{\sqrt{w}} - \frac{y w'}{2w^{3/2}} \cdot (-1) + \frac{w'}{4\sqrt{w}} \right) \right] \cdot (-1), \quad (30)$$

which is the familiar statement that $-e^{rT} \partial_K C$ equals the risk-neutral probability $\mathbb{P}(S_T > K)$ *plus* a skew correction proportional to $\partial_y w$. It is this correction that separates a digital priced with the smile from a digital priced at the Black–Scholes volatility of the same strike; see Section 8.6.

Step 2: second derivative. Differentiating (30) once more and using $\varphi'(x) = -x\varphi(x)$ together with $\partial_y d_- = -\frac{1}{\sqrt{w}} - \frac{w'}{2} \partial_w d_-$, every Gaussian term factors through $\varphi(d_-)/\sqrt{w}$. Writing $u = \frac{1}{\sqrt{w}} + \frac{w'}{2} \left(\frac{y}{w^{3/2}} - \frac{1}{2\sqrt{w}} \right)$ for the bracket in (30), one obtains after simplification

$$\partial_{KK}^2 C = \frac{e^{-rT}}{K} \frac{\varphi(d_-)}{\sqrt{w}} \left[\left(1 - \frac{y w'}{2w} \right)^2 - \frac{(w')^2}{4} \left(\frac{1}{w} + \frac{1}{4} \right) + \frac{w''}{2} \right] = \frac{e^{-rT}}{K} \frac{\varphi(d_-)}{\sqrt{w}} g(y, T).$$

The three groups of terms have a transparent origin: the squared bracket comes from differentiating d_- twice through its explicit y -dependence; the term $-\frac{(w')^2}{4} \left(\frac{1}{w} + \frac{1}{4} \right)$ from the cross terms between the explicit and the w -mediated dependence; and $\frac{w''}{2}$ from the curvature of the smile itself. Since $e^{-rT} \varphi(d_-)/(K\sqrt{w}) > 0$ always, the sign of $\partial_{KK}^2 C$ is the sign of g . Together with Theorem 3.1 this proves that butterfly-arbitrage-freeness is equivalent to $g \geq 0$, and identifies the implied density

$$p(K) = e^{rT} \partial_{KK}^2 C = \frac{g(y, T)}{K \sqrt{2\pi w}} e^{-d_-^2/2}.$$

Remark A.1. A flat smile gives $w' = w'' = 0$ and $g \equiv 1$, recovering the lognormal density. A pure linear skew $w' = \text{const}$ makes g a downward parabola in y through the term $-\frac{(w')^2}{4} \cdot \frac{y^2}{w^2} \cdot (-1)$ hidden in the squared bracket, so *every* smile that is linear in log-moneyness over an unbounded range violates $g \geq 0$ far enough out. This is the geometric content of Lee's moment formula (Lee, 2004): total variance may grow at most linearly in $|y|$, i.e. $w(y) \lesssim 2|y|$.

B Dupire in implied-variance coordinates

We derive (18) from (11). The change of variables is $(K, T) \mapsto (y, T)$ with $y = \log(K/F_T)$, so that at fixed K

$$\left. \frac{\partial y}{\partial T} \right|_K = -(r - q), \quad \left. \frac{\partial y}{\partial K} \right|_T = \frac{1}{K}.$$

Let $\tilde{C}(y, T)$ denote the call price expressed in the new variables, and introduce the normalised price $c = e^{rT} C / F_T$, which is the Black–Scholes price with $S = 1$, $K = e^y$, zero rates:

$$c(y, w) = \mathcal{N}(d_+) - e^y \mathcal{N}(d_-), \quad d_{\pm} = -\frac{y}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}.$$

The point of this normalisation is that c depends on (y, T) *only through* y and $w(y, T)$. Direct differentiation of the Black–Scholes formula gives the three identities

$$\partial_w c = \frac{\varphi(d_-) e^y}{2\sqrt{w}}, \quad \partial_y c = -e^y \mathcal{N}(d_-), \quad \partial_{yy}^2 c - \partial_y c = 2 \partial_w c \cdot \left(-\frac{1}{2}\right) \cdot (-2) = \frac{\varphi(d_-) e^y}{\sqrt{w}}, \quad (31)$$

the last of which is the statement that the Black–Scholes price solves the heat equation $\partial_w c = \frac{1}{2}(\partial_{yy}^2 c - \partial_y c)$ in the variance clock w . This is the only structural fact we need.

Numerator. At fixed K ,

$$\partial_T C|_K = e^{-rT} F_T \left[(r - q)c + \partial_y c \cdot \partial_T y|_K + \partial_w c \cdot (\partial_T w + \partial_y w \cdot \partial_T y|_K) \right] - rC + (r - q)C.$$

Substituting $\partial_T y|_K = -(r - q)$ and collecting, the terms proportional to $(r - q)$ reproduce exactly $-(r - q)K \partial_K C - qC$, so that the entire Dupire numerator reduces to

$$\partial_T C + (r - q)K \partial_K C + qC = e^{-rT} F_T \partial_w c \cdot \partial_T w, \quad (32)$$

with $\partial_T w$ taken at fixed y . This is the crucial simplification: all the rate and dividend terms cancel, and the numerator is a positive multiple of $\partial_T w$. Hence the numerator is non-negative if and only if the surface is free of calendar-spread arbitrage.

Denominator. Using $\partial_K = \frac{1}{K} \partial_y$ twice and the chain rule through w ,

$$K^2 \partial_{KK}^2 C = e^{-rT} F_T \left[\partial_{yy}^2 c - \partial_y c \right]_{\text{total}},$$

where the total second derivative expands as

$$\partial_{yy}^2 c|_{\text{total}} = \partial_{yy}^2 c + 2 \partial_{yw}^2 c \partial_y w + \partial_{ww}^2 c (\partial_y w)^2 + \partial_w c \partial_{yy}^2 w.$$

Substituting the Black–Scholes identities (31) and their consequences

$$\frac{\partial_{yw}^2 c}{\partial_w c} = \frac{1}{2} - \frac{y}{w}, \quad \frac{\partial_{ww}^2 c}{\partial_w c} = -\frac{1}{2w} - \frac{1}{8} + \frac{y^2}{2w^2},$$

all Gaussian factors cancel against $\partial_w c$ and one obtains

$$\frac{1}{2} K^2 \partial_{KK}^2 C = e^{-rT} F_T \partial_w c \cdot \left[\left(1 - \frac{y \partial_y w}{2w}\right)^2 - \frac{(\partial_y w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{\partial_{yy}^2 w}{2} \right]. \quad (33)$$

The bracket is $g(y, T)$.

Conclusion. Dividing (32) by (33), the common factor $e^{-rT} F_T \partial_w c$ — which is strictly positive — cancels, leaving

$$\sigma_{\text{loc}}^2(y, T) = \frac{\partial_T w(y, T)}{g(y, T)},$$

which is (18). Two consequences deserve emphasis. First, the denominator of Dupire’s formula in price coordinates and Durrleman’s function differ only by the strictly positive factor $e^{-rT} F_T \partial_w c$, so they vanish and change sign together; this is Corollary 4.5. Second, the formula involves w and its derivatives only, and $w = O(10^{-2})$ is a smooth, well-scaled quantity, whereas $\partial_{KK}^2 C$ is a small difference of large numbers. This is why the implied-variance form is the one to implement.

C Reproducing the results

The companion notebook `local_volatility_surface.ipynb` reproduces every number and every figure of this report from a cold start. It is self-contained: all functions are defined in the notebook itself, the only dependencies being `numpy`, `scipy`, `pandas` and `matplotlib`. Random number generation is seeded throughout, so the output is deterministic. Total runtime is a few minutes, dominated by the Monte Carlo sections.

The notebook is organised in the same order as this report: implied-volatility inversion (§2), the Heston COS pricer and the synthetic market (§7), the eSSVI calibration and the arbitrage audit (§6), the ill-posedness experiment (§5), the Dupire extraction (§4), and the Monte Carlo validation and exotics (§8).

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